The Short, the Long, and the Consequences of VAR Analysis for Structural DSGE Estimation *

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Abstract

Truncation bias could be a severe problem regarding structural VAR analysis. De Graeve and Westermark (2025) claims that the role of lag choice in SVAR is crucial to correctly identify IRFs, and they claim that choosing long lags solve the bias and, surprisingly, this reduce the variance of the point estimate. Christiano et al. (2005) is a classic paper that studies the effects of monetary policy shock in the economy. Particularly their DSGE is calibrated ex-ante using matching the model's IRFs with the ones estimated from the VAR analysis. The aim of this thesis is to perform the De Graeve and Westermark (2025) exercise, see if and how the IRFs changes, analys how the DSGE model calibration vary and if the model is consistent to this new empirical way to estimate the VARs. If so, we want to understand more about the claimed variance reduction caused by the solution of the truncation bias how works in the Frequentist and Bayesian world.

Keywords:

es. VAR, Lag Length, Truncation Bias, DSGE Estimation, Minimum Distance

JEL Codes:

es. C32, C52, E32

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Contents

1 Introduction **1**
1.1 Research Question 1
1.2 Main Contributions 1
1.3 Related Literature 1
1.4 Outline 1

2 Theoretical Framework **1**

3 Methodology **1**

4 Empirical Results **1**

5 Conclusion **1**

A Replication Files and Code **3**

B Additional Tables **3**
B.1 Implied Duration from Calvo Price Table: lags 4 8 12 3

C Additional Figures **3**
C.1 ACEL 2005 VAR Variables 4

1 Introduction

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1.3 Related Literature

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References

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (2005). Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of Political Economy*, 113(1):1–45.

De Graeve, F. and Westermark, A. (2025). Long-lag vars. *Journal of Monetary Economics*, 156.

A Replication Files and Code

Detail where the reader can find your code (e.g., a GitHub link).

B Additional Tables

B.1 Implied Duration from Calvo Price Table: lags 4 8 12

Table 1: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR ($p = 4$)**: Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

C Additional Figures

C.1 ACEL 2005 VAR Variables

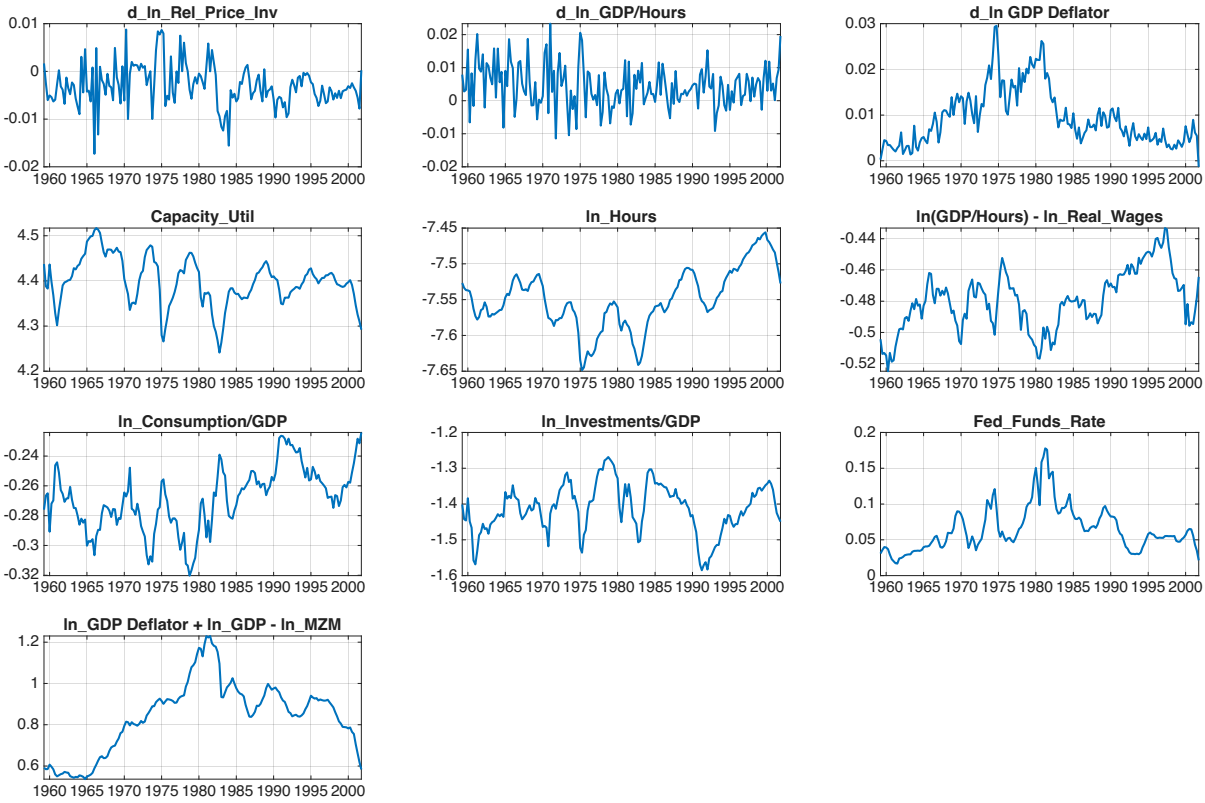


Figure 1: Raw Historical Series